EGERTON UNIVERSITY

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FACULTY OF SCIENCE

DEPARTMENT OF COMPUTER SCIENCE

COMP 493: COMPUTER SYSTEMS PROJECT

[PROJECT TITLE GOES HERE]

Submitted by:

[YOUR FULL NAME]

REG. NO: [YOUR REGISTRATION NUMBER]

Supervisor:

[SUPERVISOR'S NAME]

Coordinator:

[Coordinator's Name]

Submitted in Partial Fulfillment of the Requirements for the Award of

[DEGREE NAME, e.g., BACHELOR OF SCIENCE IN COMPUTER SCIENCE]

in the Department of Computer Science, Egerton University

[MONTH, YEAR]
[e.g., SEPTEMBER, 2025]

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DECLARATION

I, **[Student Name]**, hereby declare that this **Simple Office Automation System** project report submitted to the **Department of Computer Science, Egerton University**, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Science in Computer Science**, is my original work.

To the best of my knowledge, this project has not been submitted to any other institution or university for any academic award. All sources of information and ideas used in this project have been duly acknowledged.

I take full responsibility for the authenticity of this work and certify that any assistance received in its preparation has been properly acknowledged.

Name of Student: _____

Registration Number: _____

Signature: _____

Date: _____

Certified by Supervisor:

Name of Supervisor: _____

Signature: _____

Date: _____

ACKNOWLEDGEMENT

I wish to express my sincere gratitude to **God Almighty** ……

I am deeply grateful to my **project supervisor, [Supervisor Name]**, ----------

I would also like to extend my appreciation to the **Department of Computer Science, Egerton University**, for -------

Finally, I acknowledge --------

Thank you all.

1. PROPOSAL

PROJECT PROPOSAL

Title: Design and Implementation of a Simple Office Automation System

CHAPTER 1: INTRODUCTION

1.1 Literature Review

Office automation systems are designed to improve efficiency in office operations by automating routine tasks such as document handling, task scheduling, and communication. Studies show that office automation reduces paperwork, improves accuracy, and enhances productivity in organizations.

1.2 Problem Statement

Many offices still rely on manual processes for document management and task coordination. These methods lead to inefficiencies, delays, errors, and loss of important information.

1.3 Justification

The proposed system will automate office operations, improve efficiency, reduce paperwork, and enhance data security while minimizing operational costs.

1.4 Research Questions

1. How can office processes be automated using a simple software system?
2. What benefits does automation bring to office efficiency?
3. How secure and reliable is the proposed system?

1.5 Research Objective

To design and implement a simple office automation system that improves efficiency, accuracy, and information management.

CHAPTER 2: PROPOSED SOLUTION

2.1 Requirements Analysis

Functional requirements include user authentication, document management, task assignment, and report generation. Non-functional requirements include usability, security, reliability, and performance.

2.2 Planning

The project will follow the Software Development Life Cycle (SDLC) approach, including analysis, design, development, testing, and deployment.

2.3 Architectural Design

The system will use a client-server architecture with a web-based interface and a centralized database.

2.4 Software Development

The system will be developed using HTML, CSS, JavaScript for the frontend, PHP for backend processing, and MySQL for database management.

2.5 Testing

Testing will include unit testing, integration testing, and system testing to ensure the system meets user requirements.

2.6 Deployment

The system will be deployed on a local or cloud-based server and made accessible to authorized office staff.

CHAPTER 3: FINAL RESULTS

The final system will automate document management and task scheduling, resulting in improved efficiency and reduced errors.

CHAPTER 4: CHALLENGES, OBSTACLES, AND RISKS

Potential challenges include resistance to change, data security concerns, and limited technical skills among users.

CHAPTER 5: SCHEDULE AND BUDGET

5.1 Schedule

The project will be completed within one academic semester following an agreed timeline.

5.2 Budget

The estimated budget will cover development tools, hosting, and maintenance costs, with emphasis on open-source solutions.

CHAPTER 6: CONCLUSIONS AND FUTURE WORKS

The project demonstrates how office automation can improve efficiency. Future enhancements may include mobile integration, cloud services, and advanced reporting features.

6.1 References

Laudon, K. C., & Laudon, J. P. Management Information Systems.
Pressman, R. S. Software Engineering: A Practitioner's Approach.

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2. SOFTWARE REQUIRMENTS SPECIFICATIONS [SRS]

SYSTEM SOFTWARE REQUIRMENTS SPECIFICATIONS [SRS]

Title: Design and Implementation of a Simple Office Automation System

2.1 INTRODUCTION

2.1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements of the Simple Office Automation System. This document acts as a guide for developers, supervisors, and users involved in the project.

2.1.2 Document Conventions

This document follows standard software engineering conventions. Section headings are numbered, and technical terms are used consistently throughout the document.

2.1.3 Intended Audience and Reading Suggestions

The intended audience includes project supervisors, system developers, testers, and end users. It is recommended that readers read the entire document to fully understand system requirements.

2.1.4 Scope

The Simple Office Automation System is designed to automate routine office activities such as document management, task assignment, scheduling, and report generation.

2.2 OVERALL DESCRIPTION

2.2.1 Product Perspective

The system will operate as a standalone web-based application accessible through a web browser.

2.2.2 Product Features

- Secure user login and authentication
- Document upload, storage, and retrieval

- Task assignment and monitoring
- Automated reporting

2.2.3 User Problem Statement

Manual office processes are time-consuming and prone to errors, resulting in inefficiencies and data loss.

2.2.4 User Objectives

- Improve efficiency in office operations
- Reduce paperwork
- Improve data accuracy and accessibility

2.2.5 Operating Environment

The system will run on modern web browsers and will be hosted on a server running Windows or Linux.

2.2.6 User Documentation

User manuals and online help documentation will be provided to assist users in system operation.

2.2.7 Assumptions and Dependencies

Users are assumed to have basic computer skills and access to internet-enabled devices.

2.3 SYSTEM FEATURES

1. User Authentication
2. Document Management
3. Task Management
4. Report Generation

2.4 EXTERNAL INTERFACE REQUIREMENTS

2.4.1 User Interfaces

The system shall provide a web-based graphical user interface that is easy to navigate.

2.4.2 Hardware Interfaces

The system shall operate on standard computing devices such as desktops, laptops, tablets, and smartphones.

2.4.3 Software Interfaces

The system shall interface with a MySQL database and a PHP-based web server.

2.4.4 Communication Interfaces

The system shall use HTTP and HTTPS protocols for communication.

2.5 NON-FUNCTIONAL REQUIREMENTS

2.5.1 Safety

The system shall include data backup and recovery mechanisms.

2.5.2 Performance

The system shall respond to user requests within acceptable response times.

2.5.3 Security

The system shall enforce authentication, authorization, and data protection mechanisms.

2.5.4 Software Quality Attributes

The system shall be reliable, maintainable, scalable, and user-friendly.

2.6 PRELIMINARY OBJECT ORIENTED DOMAIN ANALYSIS

The main objects in the system include **User**, **Administrator**, **Document**, **Task**, and **Report**. These objects interact to support office automation operations.

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3. SOFTWARE DESIGN DOCUMENTATION (SDD)

SYSTEM SOFTWARE DESIGN DOCUMENTATION

Title: Design and Implementation of a Simple Office Automation System

3.1 INTRODUCTION

3.1.1 Purpose and Scope

The purpose of this Software Design Documentation (SDD) is to describe the design and architecture of the Simple Office Automation System. This document provides detailed design information required to implement the system based on the approved requirements.

3.1.2 Project Executive Summary

The Simple Office Automation System is designed to automate routine office operations such as document management, task assignment, and reporting. The system aims to improve efficiency, reduce paperwork, and enhance information management through a web-based solution.

3.1.3 Document Organization

This document is organized into sections describing system architecture, database design, user interfaces, detailed component design, and system integrity controls.

3.1.4 Points of Contact

- **Student:** _____
- **Registration Number:** _____
- **Supervisor:** _____
- **Department:** Computer Science, Egerton University

3.1.5 Glossary

- **SDD:** Software Design Documentation
- **UI:** User Interface
- **DB:** Database
- **HMI:** Human–Machine Interface

3.2 SYSTEM ARCHITECTURE

3.2.1 Hardware Architecture

The system will operate on standard computing devices such as desktop computers, laptops, tablets, and smartphones. The server hardware will host the application and database.

3.2.2 Software Architecture

The system adopts a **client-server architecture**:

- **Client Layer:** Web browser (User Interface)
- **Application Layer:** Web server with backend logic
- **Data Layer:** MySQL database

3.3 FILE AND DATABASE DESIGN

3.3.1 File Design

- User files: authentication data
- Document files: uploaded office documents
- Log files: system activity records

3.3.2 Database Design

Key database tables include:

- **Users:** stores user credentials and roles
- **Documents:** stores document details and metadata
- **Tasks:** stores task assignments and status
- **Reports:** stores generated reports

Each table has a primary key to uniquely identify records.

3.4 HUMAN–MACHINE INTERFACE (HMI)

3.4.1 Inputs

- User login credentials
- Document uploads
- Task creation and updates

3.4.2 Outputs

- Confirmation messages

- Task status reports
- Document retrieval results
- Summary and activity reports

The interface is designed to be user-friendly, simple, and intuitive.

3.5 DETAILED DESIGN

3.5.1 Hardware Design

The system requires basic computing devices with internet access and a server for hosting the application.

3.5.2 Software Design

The system is developed using:

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** PHP
- **Database:** MySQL

3.5.3 Support Design

System support includes user training, documentation, and periodic system maintenance.

3.5.4 Final Communication Design

The system uses HTTP/HTTPS protocols for communication between the client and server to ensure secure data transmission.

3.6 SYSTEM INTEGRITY CONTROLS

The system includes integrity controls to ensure data accuracy, security, and reliability:

- User authentication and authorization
- Input validation
- Data backup and recovery
- Access control based on user roles
- Audit logs for system activities

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4. TEST PLAN

SYSTEM TEST PLAN

Title: Design and Implementation of a Simple Office Automation System

4.1 INTRODUCTION

4.1.1 Goals and Objectives

The goal of this test plan is to ensure that the Simple Office Automation System meets the functional and non-functional requirements as defined in the SRS. Objectives include:

- Verifying system functionality
- Identifying and fixing defects
- Ensuring usability, reliability, and performance

4.1.2 Scope of Statement

The scope of testing includes:

- User authentication and authorization
- Document management (upload, retrieval, deletion)
- Task assignment and tracking
- Report generation
- System performance and security checks

4.2 TEST PLAN

4.2.1 Testing Strategy

Testing will follow a **structured approach**, including:

- **Unit Testing:** Testing individual components (e.g., login module, document upload)
- **Integration Testing:** Ensuring components work together correctly
- **System Testing:** Validating end-to-end system functionality
- **Acceptance Testing:** Confirming the system meets user requirements

4.2.2 Interfaces

Testing will cover:

- User interfaces (web pages)
- Database interfaces (CRUD operations)
- Communication interfaces (HTTP/HTTPS)

4.2.3 Bug Triage

Identified defects will be logged, categorized by severity (critical, major, minor), and addressed based on priority. A bug tracking tool (e.g., Jira, Trello) may be used.

4.3 TEST PROCEDURE

4.3.1 Testing Criteria

- **Success criteria:** System functions correctly as specified
- **Failure criteria:** System fails to meet functional or non-functional requirements

4.3.2 Testing Procedure

1. **Unit Testing:** Test individual modules (login, document upload, task management)
2. **Integration Testing:** Test interaction between modules (e.g., document linked to tasks)
3. **System Testing:** Test complete workflow from user login to report generation
4. **Acceptance Testing:** Conducted with selected users to validate usability and performance

4.4 TESTING RESOURCE PLANNING

4.4.1 System Resources

- Web server (Windows/Linux)
- Client computers, laptops, tablets
- Database server (MySQL)

4.4.2 Human Resources

- Test engineer / student developer
- Project supervisor
- Selected end users for acceptance testing

4.5 TEST ENVIRONMENT

The system will be tested in a controlled environment:

- Web server running Apache or Nginx
- Database server (MySQL)
- Web browsers: Chrome, Firefox
- Stable network connectivity

4.6 TEST SCHEDULE AND ESTIMATION

Phase	Duration (days)	Responsible Person	Notes
Unit Testing	xx	Developer	xx
Integration Testing	xx	xx	xx
System Testing	xx	xx	xx
Acceptance Testing	xx	xx	Verify user satisfaction

Total estimated testing time: **XXXXX duration**

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5. USER MANUAL

SYSTEM USER MANUAL

Title: Design and Implementation of a Simple Office Automation System

5.1 SYSTEM OVERVIEW

The Simple Office Automation System is a web-based application designed to automate routine office operations, including:

- Document management (uploading, storing, retrieving, and deleting documents)
- Task management (assignment, tracking, and completion status)
- Report generation (summary and activity reports)
- User management (login, roles, and permissions)

The system aims to improve efficiency, reduce paperwork, enhance data security, and facilitate smooth communication among office staff.

5.2 SYSTEM REQUIREMENTS

5.2.1 Hardware Requirements

- Desktop or laptop computer with minimum 4GB RAM
- Tablet or smartphone with internet access
- Web server for hosting the application
- Database server (MySQL)

5.2.2 Software Requirements

- Web browser (Google Chrome, Mozilla Firefox)
- PHP server (e.g., Apache or Nginx)
- MySQL database server
- Internet connectivity

5.3 SYSTEM FEATURES

1. **User Authentication:**
    - Secure login with username and password
    - Role-based access control
2. **Document Management:**
    - Upload, download, view, edit, and delete documents
    - Organize documents by categories
3. **Task Management:**
    - Assign tasks to staff members
    - Track task progress and status

    - Mark tasks as completed
4. **Reporting:**
    - Generate reports for documents and tasks
    - Export reports in PDF format
5. **Administrative Features:**
    - Add, update, or delete users
    - Assign user roles and permissions

5.4 SYSTEM SUPPORT

- **User Support:** User manuals and online help are available to assist with system navigation.
- **Technical Support:** System maintenance, updates, and troubleshooting are provided by IT staff or the project supervisor.
- **Training:** Users will receive brief training sessions on system usage and best practices.